

Principles of Information Security

End Semester Examination Question Bank

- Exercise 1** Prove that no byzantine agreement protocol exists for 3 parties with 1 among them being byzantine.
- Exercise 2** State the authenticated byzantine agreement protocol and show how byzantine protocol can be reduced to broadcast problem.
- Exercise 3** State and prove No-Cloning theorem.
- Exercise 4** Explain how Bitcoin protocol achieves the 4 fundamental goals discussed in class.
- Exercise 5** State and prove Chinese Remainder Theorem.
- Exercise 6** State and prove Fermat's Little Theorem.
- Exercise 7** What are probabilistically checkable proofs? Using Collision Resistant Hash Functions and PCPs, can you suggest a scheme that requires only $O(\log n)$ bits of communication to verify a NP problem?
HINT: Try use Merkle Damgard Tree Construction. This is called Succinct Argument
- Exercise 8** Explain a protocol for performing addition and multiplication for SMPC.
- Exercise 9** Explain the BB84 protocol for Quantum Key Distribution. How do Alice and Bob figure out if an eavesdropper is trying to listen to the channel or not?
- Exercise 10** What is the birthday attack? Prove that the probability of collision of the birthday attack scheme > 0.5 .
- Exercise 11** Prove formally that any coalition of fewer than t participants learns no information about the secret in Shamir's scheme. State precisely the probability space you are working in and identify the independence you must show.
- Exercise 12** Let p be a prime with $n < p$. Show that there is exactly one polynomial of degree $< t$ that interpolates a given set of t shares; then argue why uniqueness fails if the degree bound is relaxed to $\leq t$.
- Exercise 13** Demonstrate that Shamir's scheme attains the minimal share length ($\log p$ bits) for information-theoretic threshold schemes. Outline why any lossless scheme with perfect privacy must have shares at least as long as the secret.

- Exercise 14** Explain why Shamir’s scheme has zero reconstruction error, then design a modification that allows $\epsilon > 0$ reconstruction error but reduces expected share size. Analyze the trade-off.
- Exercise 15** Over the field F_{19} , generate a $(3, 5)$ secret sharing scheme for the secret $s = 7$. Publish all five shares. Then show step-by-step how any three specific shares reconstruct s using Lagrange interpolation.
- Exercise 16** Suppose one participant forges their share during reconstruction in a (t, n) scheme. Give an algorithm that detects cheating when an authenticated channel is available only between the dealer and each participant (not between participants). Prove correctness.
- Exercise 17** In some organizations a “director” counts as two votes. Design and analyze a weighted-threshold version of Shamir’s scheme in which each player i has weight w_i and the secret is recoverable iff the total weight of cooperating players $\geq T$. Show how to assign each player an appropriate number of shares without increasing total share size asymptotically.
- Exercise 18** Compare the security of Shamir’s scheme when the adversary chooses which shares to corrupt before vs. during the protocol execution. Does the scheme remain perfectly private in both cases? Justify.
- Exercise 19** Let $n > 3t$ parties hold Shamir shares of a polynomial $g(x)$ of degree at most $2t$. Give a formal proof that the GRR protocol outputs shares of a polynomial $f(x)$ of degree at most t with $f(0) = g(0)$ and such that the joint distribution of $\langle f(x_i) \rangle_i$ is uniform among all valid sharings of that secret.
- Exercise 20** The original GRR degree reduction protocol uses two rounds of degree- t Shamir re-sharings. Propose a one-round variant that first locally blinds each share with a degree- t random polynomial and then performs a single reconstruction. Analyse whether the new protocol still achieves perfect privacy and robustness.
- Exercise 21** Construct an explicit adversary that breaks either privacy or correctness of GRR when $t = \lfloor n/3 \rfloor$. Hint: try to make two honest subsets interpolate two different secrets that both satisfy the degree- $2t$ constraints.
- Exercise 22** Show that a perfectly secure 1-out-of-2 Oblivious Transfer primitive is impossible against an all-powerful adversary that can corrupt one party. Conclude why the standard GMW protocol can only aim for computational (or statistical) security unless your model assumes a trusted setup such as pre-shared OT correlations.
- Exercise 23** Design a three-party $n = 3, t = 1$ protocol that evaluates a single AND gate with perfect security using only additive secret sharing over F_2 and pairwise private channels (no OT). Prove both privacy and correctness or explain why this is impossible.

- Exercise 24** The original GMW protocol achieves statistical—but not perfect—security because of the use of OT. Design a perfectly secure variant for Boolean AND gates when at most one party is dishonest and the OT primitive itself is replaced by a 1-out-of-2 perfect secret sharing of the truth table. Analyze round and communication costs.
- Exercise 25** Show that OT is complete for secure two-party computation: i.e., any PPT functionality f can be computed securely from black-box access to OT and local computation. Outline the reduction and specify the exact number of OT calls per AND gate in the resulting GMW-style protocol.
- Exercise 26** Give a concrete example (with explicit distributions) in which an OT protocol is statistically—but not perfectly—private. Quantify the distinguishing advantage and explain why it cannot be driven to 0 without increasing communication.
- Exercise 27** Give a protocol that implements 1-out-of- n OT using $O(\log n)$ instances of 1-out-of-2 OT and symmetric encryption. Prove both correctness and privacy.
- Exercise 28** Show that any perfectly private protocol for equality ($x = y$) over k -bit strings needs to transmit at least k bits in total.
- Exercise 29** Prove that no two-message protocol can securely compute the AND functionality with malicious security in the plain model, assuming only one-way functions.
- Exercise 30** State and prove Shor's Algorithm (proof-sketch) and discuss its implications on public key cryptography.